

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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Section: CSM-B

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1. TITLE

HTML5 Canvas Drawing and CSS Web Page Styling

2. AIM

To draw basic shapes and text using the HTML5 Canvas API, and to design a visually styled webpage using CSS properties such as gradients, box-shadow, flexbox, and hover effects.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the HTML5 <canvas> element and its 2D drawing context.
- To draw geometric shapes (rectangle, circle, line) and text on a canvas using JavaScript.
- To apply CSS styling techniques such as gradients, shadows, and border-radius.
- To design a responsive card layout using Flexbox.
- To implement hover effects on navigation links and buttons using CSS.

4. THEORY

The HTML5 <canvas> element provides a drawing surface that can be manipulated using JavaScript through its 2D rendering context. Methods such as fillRect(), arc(), moveTo(), lineTo(), stroke(), and fillText() allow shapes and text to be drawn programmatically. CSS (Cascading Style Sheets) is used to control the presentation of HTML elements — properties like background (gradients), box-shadow, border-radius, and the Flexbox layout model (display: flex) enable modern, responsive, and visually appealing page designs. Pseudo-classes such as :hover allow interactive styling in response to user actions.

5. ALGORITHM / PROCEDURE

1. Create an HTML file and add a <canvas> element with defined width and height.
2. Access the canvas using getElementById() and obtain its 2D rendering context.
3. Use fillRect() to draw a filled rectangle.
4. Use beginPath(), arc(), and fill() to draw a filled circle.
5. Use beginPath(), moveTo(), lineTo(), and stroke() to draw a line.

6. Use font, fillStyle, and fillText() to display text on the canvas.
7. Create a second HTML file and define CSS rules in a <style> block.
8. Style the header, navigation bar, and add a hover effect on nav links.
9. Create a card container using Flexbox and style individual cards.
10. Style buttons with a hover effect and add a footer.
11. Test and verify both pages in a browser.

6. CODE EXPLANATION

The first program (code1) uses the HTML5 Canvas API to draw a blue rectangle with fillRect(), a red circle with arc() and fill(), a black diagonal line with moveTo(), lineTo() and stroke(), and finally displays the student's name as text using fillText(). The second program (code2) is a CSS-styled webpage: the body uses a linear-gradient background, the header has a dark blue background with a text-shadow, the navigation bar uses flex-friendly inline links with a :hover color and background change, and three content cards are laid out side by side using display: flex inside a .container class, each styled with padding, border-radius, and box-shadow. Buttons within the cards also have a :hover effect, and the footer is styled with a dark background and centered white text.

7. PROGRAM

code1.html — HTML5 Canvas Drawing

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML5 Canvas</title>

  <style>
    body {
      text-align: center;
      margin-top: 30px;
      font-family: Arial, sans-serif;
    }

    h2 {
      text-align: center;
    }

    canvas {
      border: 2px solid black;
      display: block;
      margin: auto;
    }
  </style>
</head>

<body>

  <h2>HTML5 Canvas</h2>
```

```
<!-- Canvas Area -->
<canvas id="myCanvas" width="500" height="300"></canvas>
```

```
<script>
```

```
    // Get canvas
    var c = document.getElementById("myCanvas");
```

```
    // Get drawing tool
    var ctx = c.getContext("2d");
```

```
    // Draw Rectangle
    ctx.fillStyle = "blue";
    ctx.fillRect(20, 20, 100, 60);
```

```
    // Draw Circle
    ctx.beginPath();
    ctx.arc(200, 50, 30, 0, 2 * Math.PI);
    ctx.fillStyle = "red";
    ctx.fill();
```

```
    // Draw Line
    ctx.beginPath();
    ctx.moveTo(280, 20);
    ctx.lineTo(400, 100);
    ctx.strokeStyle = "black";
    ctx.lineWidth = 2;
    ctx.stroke();
```

```
    // Display Name
    ctx.font = "25px Arial";
    ctx.fillStyle="black"
    ctx.fillText("Rama Karri", 150, 200);
```

```
</script>
```

```
</body>
</html>
```

code2.html — CSS Styled Webpage

```
<!DOCTYPE html>
<html>
```

```
<head>
<meta charset="UTF-8">
```

```
<title>My Styled Webpage</title>

<style>

/* Body Styling */
body {
    margin: 0;
    font-family: Arial, sans-serif;
    background: linear-gradient(to right, #74ebd5, #acb6e5);
}

/* Header Styling */
header {
    background-color: darkblue;
    color: white;
    text-align: center;
    padding: 25px;
    text-shadow: 2px 2px 5px black;
}

header h1 {
    font-family: Georgia, serif;
    font-size: 40px;
}

/* Navigation Bar */
nav {
    background-color: black;
    padding: 15px;
    text-align: center;
}

/* Navigation Links */
nav a {
    color: white;
    text-decoration: none;
    font-size: 18px;
    margin: 20px;
    padding: 5px;
}

/* Navigation Hover Effect */
nav a:hover {
    color: yellow;
    background-color: green;
    border-radius: 5px;
}

/* Card Container */
.container {
    display: flex;
    flex-direction: row;
```

```
    justify-content: center;
    gap: 20px;
    margin: 30px;
}

/* Card Design */
.card {
    background-color: white;
    width: 250px;
    padding: 20px;
    border: 2px solid black;
    border-radius: 15px;
    box-shadow: 5px 5px 10px gray;
}

/* Card Heading */
.card h2 {
    color: darkblue;
    font-family: Georgia, serif;
    font-size: 25px;
}

/* Card Paragraph */
.card p {
    font-family: Arial;
    font-size: 16px;
}

/* Button Styling */
button {
    background-color: darkblue;
    color: white;
    padding: 10px 20px;
    border: none;
    border-radius: 8px;
}

/* Button Hover Effect */
button:hover {
    background-color: red;
    cursor: pointer;
}

/* Footer Styling */
footer {
    background-color: black;
    color: white;
    text-align: center;
    padding: 20px;
    margin-top: 30px;
}
```

```
</style>

</head>

<body>

<!-- Header -->
<header>
  <h1>My Styled Webpage</h1>
  <p>HTML5 and CSS Web Styling</p>
</header>

<!-- Navigation Bar -->
<nav>
  <a href="#">Home</a>
  <a href="#">About</a>
  <a href="#">Services</a>
  <a href="#">Contact</a>
</nav>

<!-- Three Content Cards -->
<div class="container">

  <div class="card">
    <h2>HTML5</h2>
    <p>HTML5 is used to create the structure of webpages.</p>
    <button>Read More</button>
  </div>

  <div class="card">
    <h2>CSS</h2>
    <p>CSS is used to design webpages with colors and effects.</p>
    <button>Explore</button>
  </div>

  <div class="card">
    <h2>JavaScript</h2>
    <p>JavaScript adds interaction and dynamic features.</p>
    <button>Learn</button>
  </div>
</div>

<!-- Footer -->
<footer>
  <p>© 2026 My Styled Webpage</p>
</footer>
```

</body>

</html>

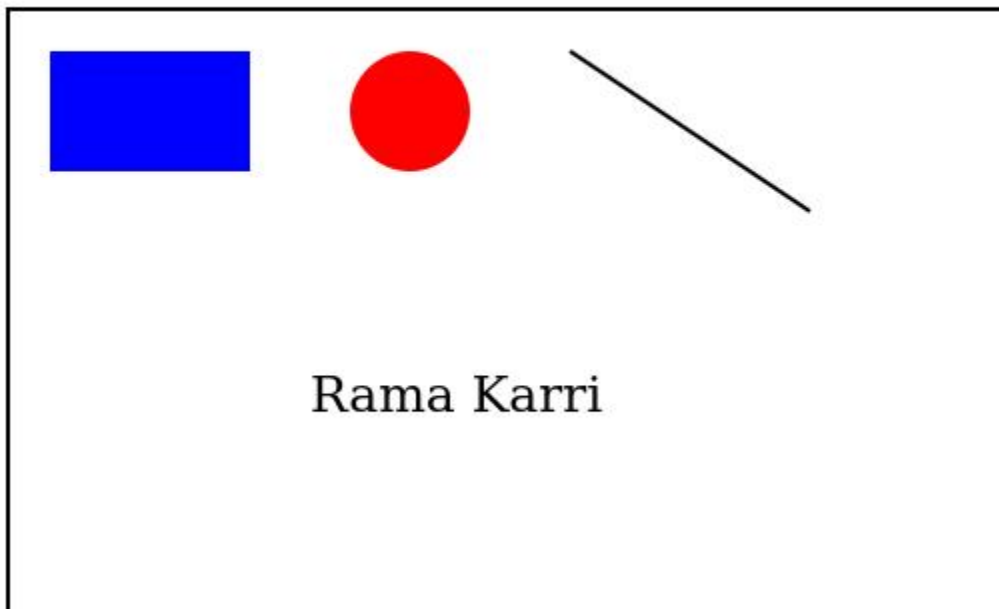
8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Canvas drawing (code1)	Verify rectangle, circle, line and text render on canvas	All shapes and text rendered correctly	Pass
TC-02	CSS styled webpage (code2)	Verify header, nav, cards, hover effects and footer styling	Page rendered with gradient background, styled cards and footer	Pass

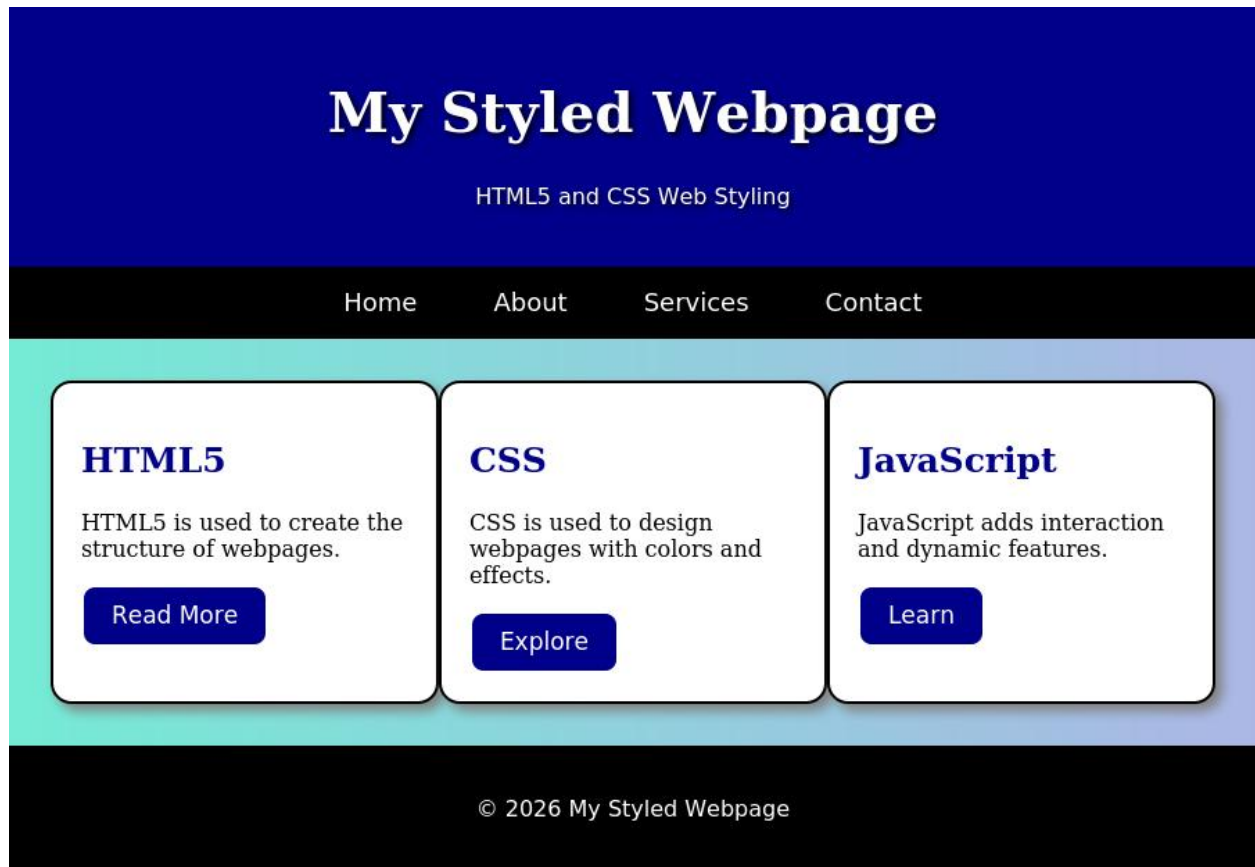
9. OUTPUT

Output of code1.html:

HTML5 Canvas



Output of code2.html:



10. RESULT

Thus, basic shapes and text were successfully drawn using the HTML5 Canvas API, and a visually styled webpage using CSS gradients, Flexbox layout, box-shadow, and hover effects was designed and verified in a browser.